LAB EXPERIMENTS

N.Yugesh(192424329)

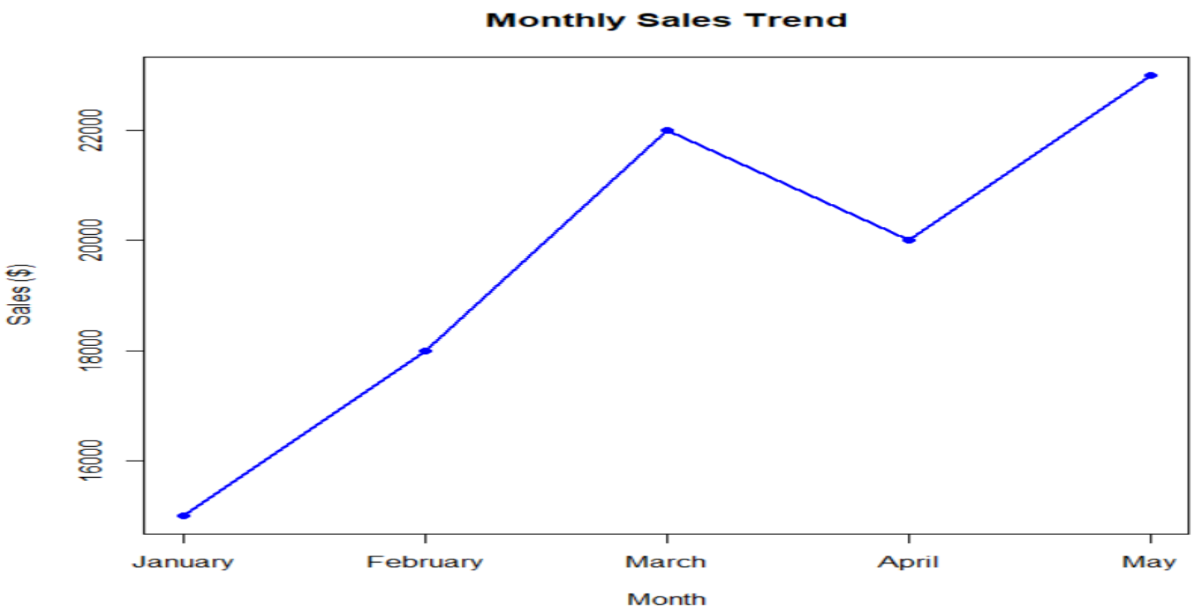
Experiment 1 – Set 1: Monthly Sales Data

1. Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

Code:

```
Month <- c("January","February","March","April","May")
Sales <- c(15000,18000,22000,20000,23000)
plot(
Sales,
type="o",
col="blue",
xaxt="n",
xlab="Month",
ylab="Sales ($)",
main="Monthly Sales Data"
)
axis(1,at=1:5,labels=Month)
```

output:

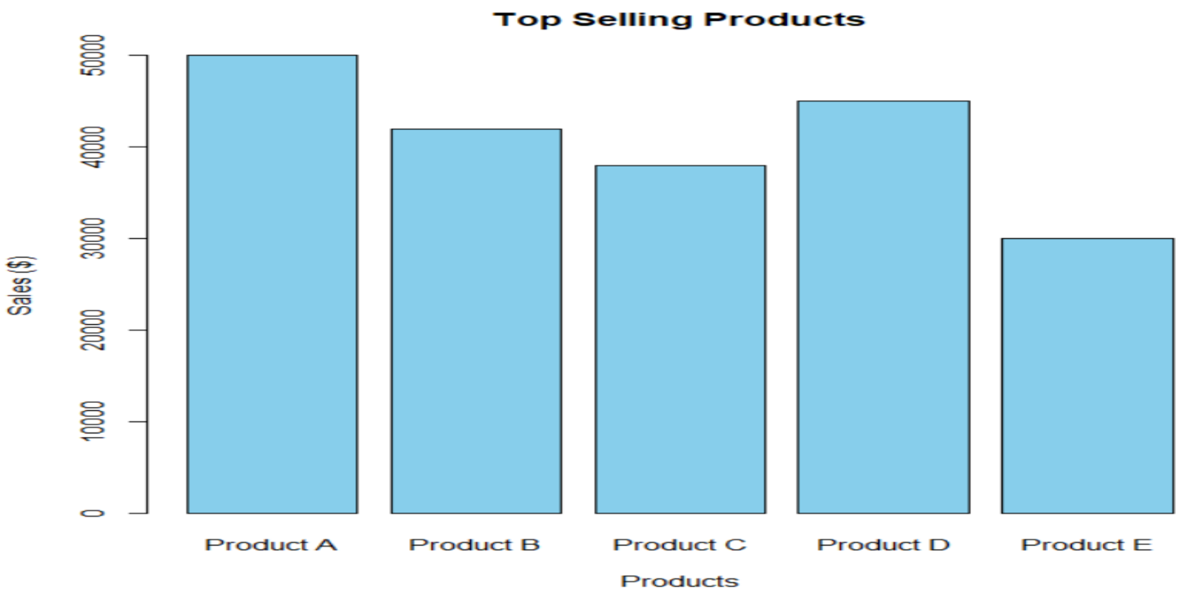


2. Generate a bar chart to display the top-selling products for the year. Label the chart elements.

Code:

```
Product <- c("Product A","Product B","Product C","Product D","Product E")
Sales <- c(2500,4200,3100,5000,3800)
barplot(
Sales,
names.arg=Product,
col="orange",
xlab="Products",
ylab="Sales",
main="Top Selling Products"
)
```

Output:



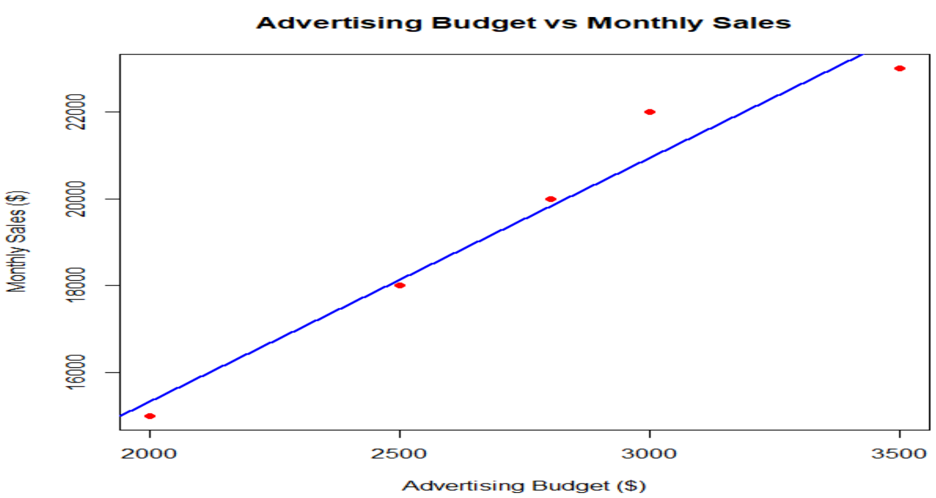
3. Develop a scatter plot to explore the relationship between advertising budget and monthly sales.

Explain the insights drawn from the scatter plot.

Code:

```
Advertising <- c(2000,2500,3000,2800,3500)
Sales <- c(15000,18000,22000,20000,23000)
plot(
Advertising,
Sales,
pch=19,
col="red",
xlab="Advertising Budget ($)",
ylab="Monthly Sales ($)",
main="Advertising Budget vs Monthly Sales"
)
abline(lm(Sales~Advertising),col="blue",lwd=2)
```

output:



SET 2 – Customer Feedback Analysis

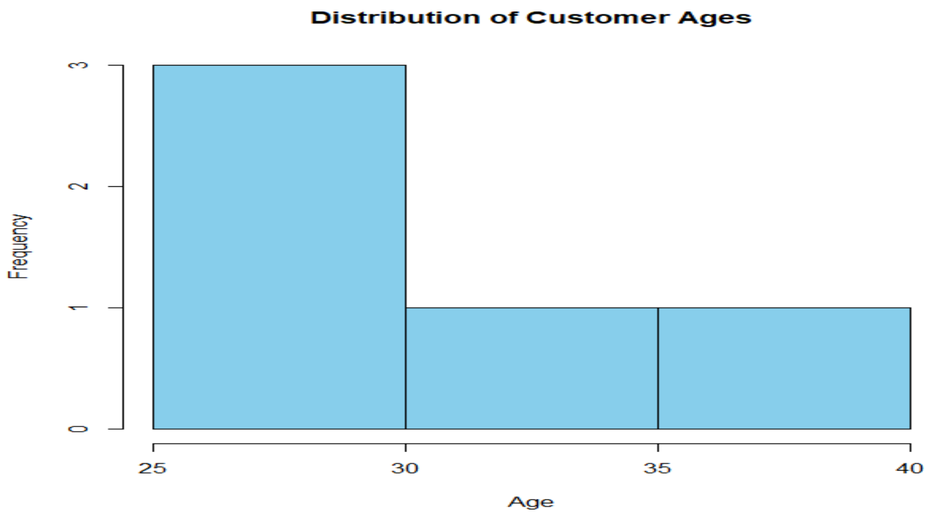
1. Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

Code:

```
Age <- c(25,30,35,28,40)

hist(
Age,
col="skyblue",
main="Distribution of Customer Ages",
xlab="Age",
ylab="Frequency"
)
```

Output:



2. Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

Code:

```
Satisfaction <- c(4,5,3,4,5)

table_score <- table(Satisfaction)
pie(
table_score,
main="Customer Satisfaction Scores",
col=c("red","green","yellow")
)
```

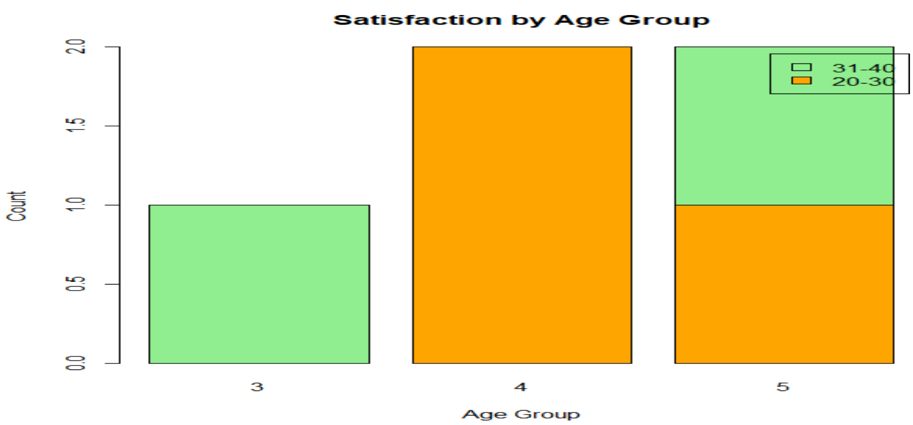
Output:

3. Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

Code:

```
AgeGroup <- c("20-30","20-30","31-40","20-30","31-40")
Satisfaction <- c(4,5,3,4,5)
data <- table(AgeGroup,Satisfaction)
barplot(
data,
col=c("orange","lightgreen","pink"),
main="Satisfaction by Age Group",
xlab="Age Group",
ylab="Count",
legend=TRUE
)
```

Output:



3. Develop a word cloud from open-ended customer feedback to identify prevalent customer

Code:

```
install.packages("wordcloud")
library(wordcloud)
feedback <- c(
"Good Service",
"Excellent Support",
"Fast Delivery",
"Good Quality",
"Excellent Product"
)
wordcloud(
feedback,
colors=rainbow(5)
)
```

Output:

Set 3: Employee Performance Evaluation

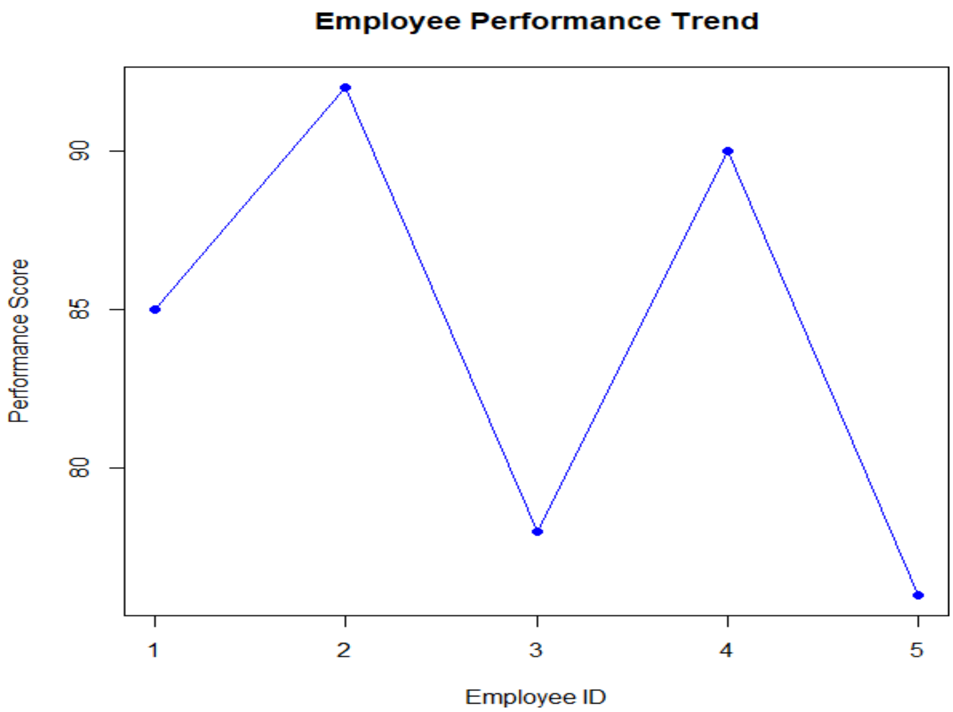
1. Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.

Code:

```
Employee_ID <- c(1,2,3,4,5)
Performance_Score <- c(85,92,78,90,76)

plot(
  Employee_ID,
  Performance_Score,
  type="o",
  col="blue",
  pch=16,
  xlab="Employee ID",
  ylab="Performance Score",
  main="Employee Performance Trend"
)
```

Output:



2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

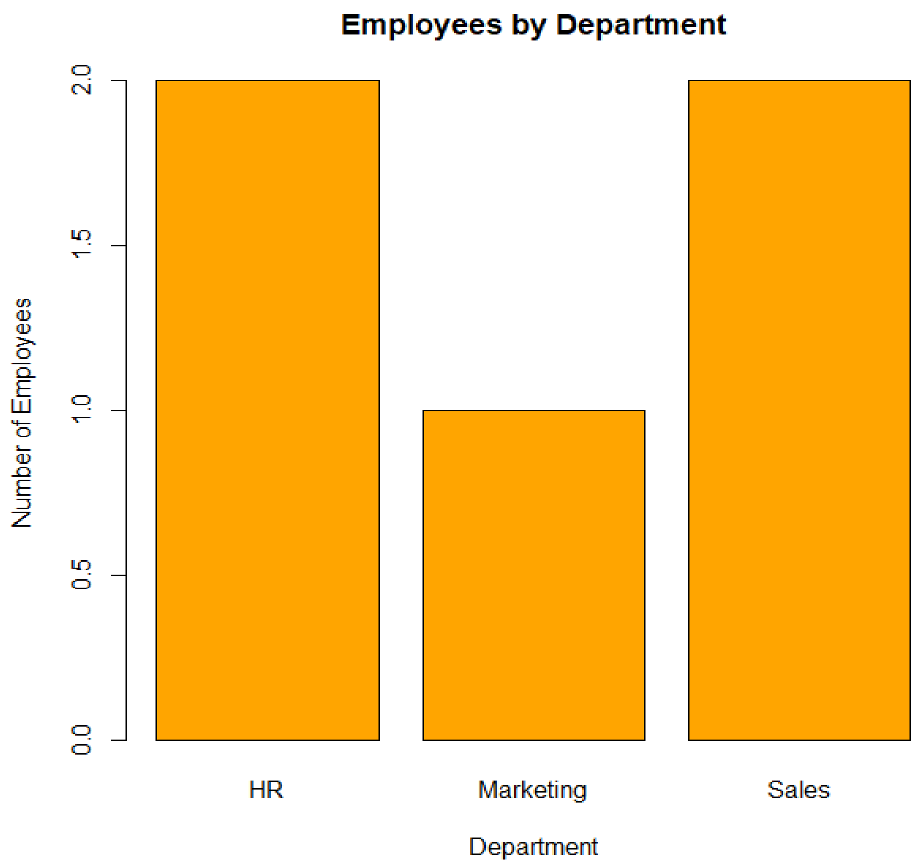
Code:

```
Department <- c("Sales","HR","Marketing","Sales","HR")

dept_count <- table(Department)

barplot(
  dept_count,
  col="orange",
  main="Employees by Department",
  xlab="Department",
  ylab="Number of Employees"
)
```

Code:



3. Build a scatter plot to analyze the correlation between years of service and performance scores.

Explain any insights.

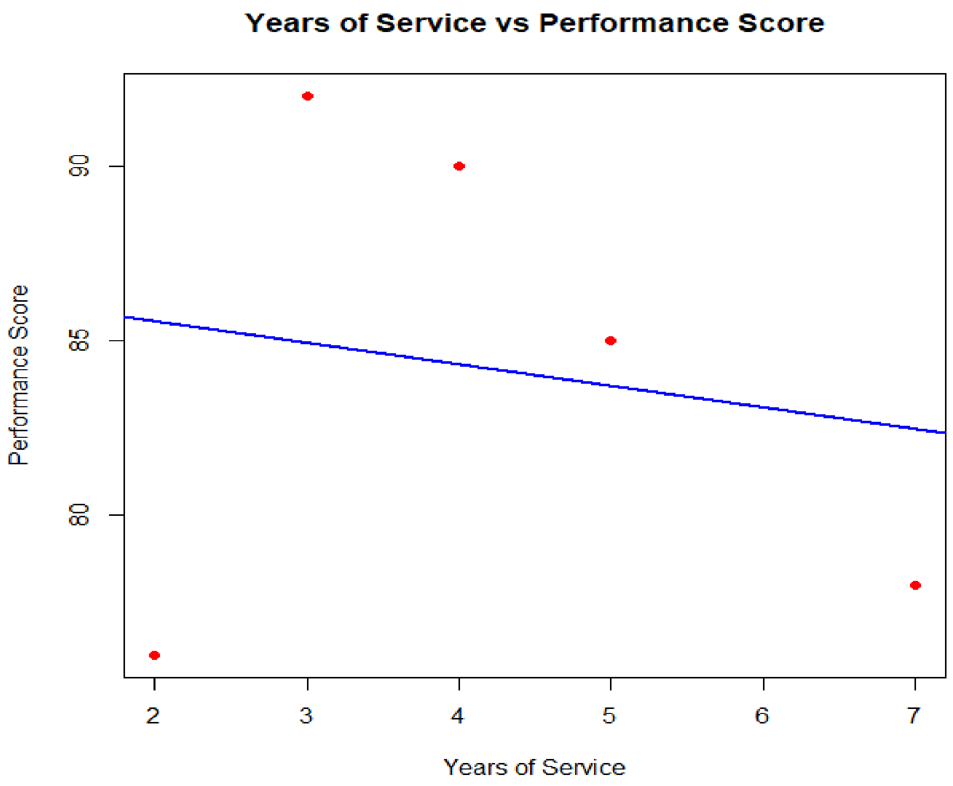
Code:

```
Years_of_Service <- c(5,3,7,4,2)
Performance_Score <- c(85,92,78,90,76)

plot(
  Years_of_Service,
  Performance_Score,
  pch=19,
  col="red",
  xlab="Years of Service",
  ylab="Performance Score",
  main="Years of Service vs Performance Score"
)

abline(lm(Performance_Score ~ Years_of_Service),
       col="blue",
       lwd=2)
```

output:



Set 4: Product Inventory Management

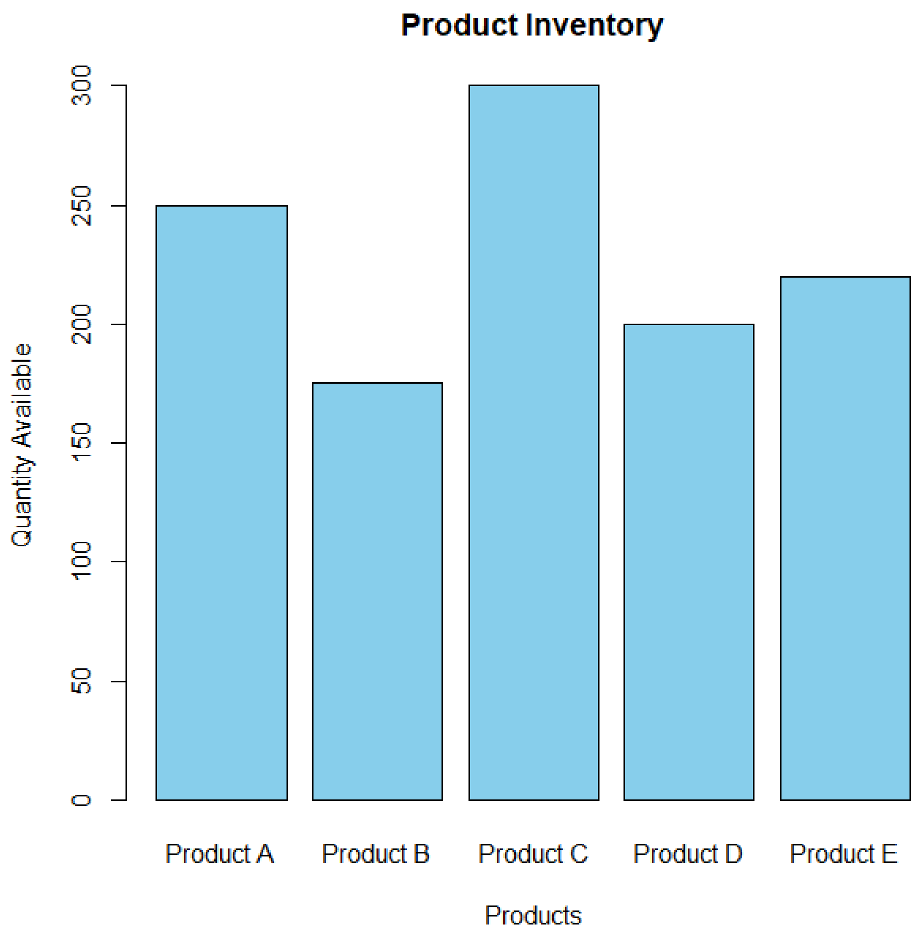
1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

Code:

```
Product <- c("Product A","Product B","Product C","Product D","Product E")
Quantity <- c(250,175,300,200,220)

barplot(
  Quantity,
  names.arg = Product,
  col = "skyblue",
  xlab = "Products",
  ylab = "Quantity Available",
  main = "Product Inventory"
)
```

Output:



2. Generate a stacked bar chart to show the quantity of each product within different product categories.

Code:

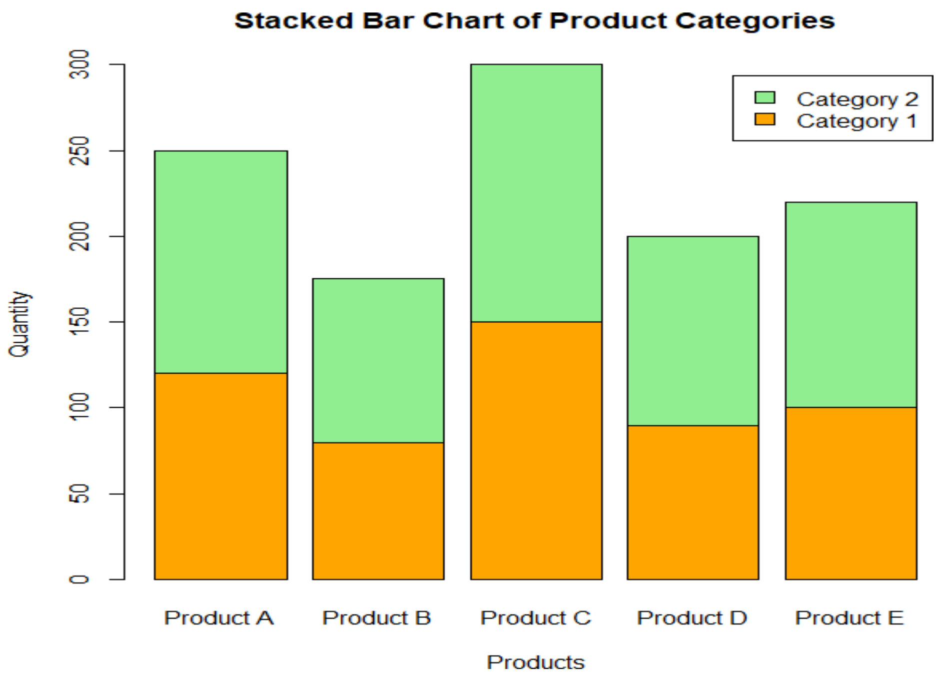
```
Category <- matrix(
c(120,130,
80,95,
150,150,
90,110,
100,120),
nrow=2)

colnames(Category) <- c("Product A","Product B","Product C","Product
D","Product E")

rownames(Category) <- c("Category 1","Category 2")

barplot(
Category,
col=c("orange","lightgreen"),
main="Stacked Bar Chart of Product Categories",
xlab="Products",
ylab="Quantity",
legend=rownames(Category)
)
```

Output:



3. Build a scatter plot to explore the relationship between product price and quantity available.

Explain the findings.

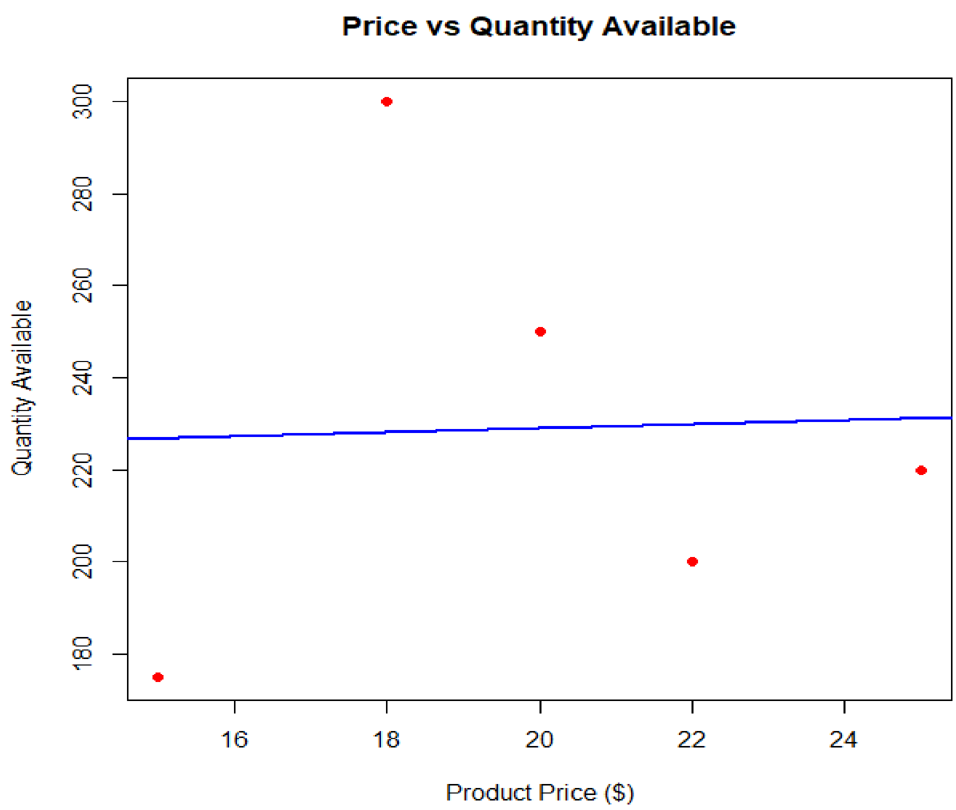
Code:

```
Price <- c(20,15,18,22,25)
Quantity <- c(250,175,300,200,220)

plot(
Price,
Quantity,
pch=19,
col="red",
xlab="Product Price ($)",
ylab="Quantity Available",
main="Price vs Quantity Available"
)

abline(lm(Quantity~Price),col="blue",lwd=2)
```

output:



Set 5: Website Analytics

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

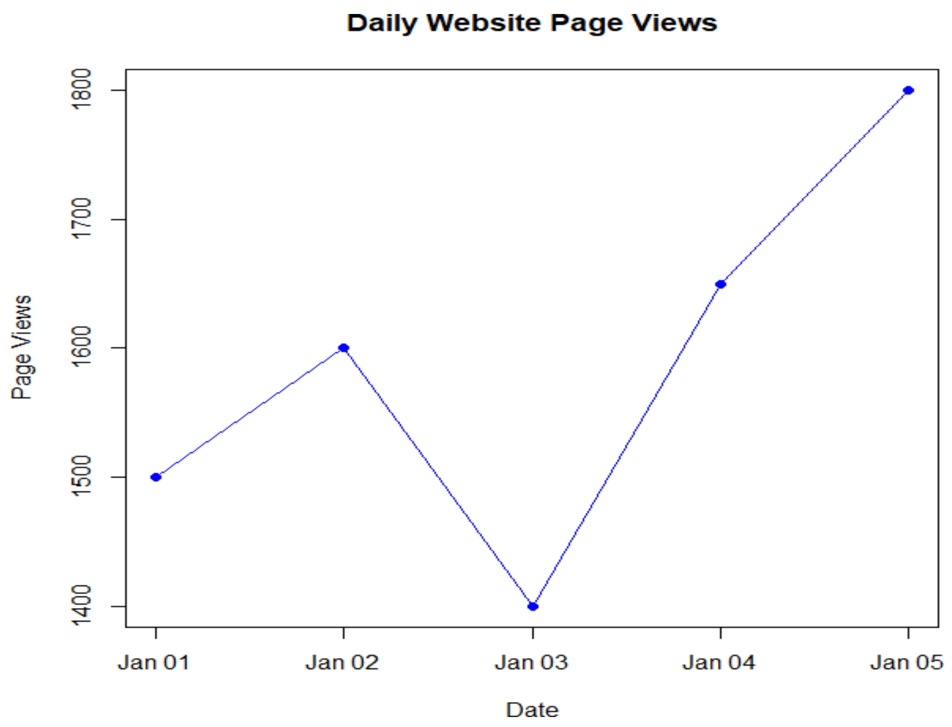
Code:

```
Date <- as.Date(c("2023-01-01",
                  "2023-01-02",
                  "2023-01-03",
                  "2023-01-04",
                  "2023-01-05"))

Page_Views <- c(1500,1600,1400,1650,1800)

plot(
Date,
Page_Views,
type="o",
col="blue",
pch=16,
xlab="Date",
ylab="Page Views",
main="Daily Website Page Views"
)
```

Output:



2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

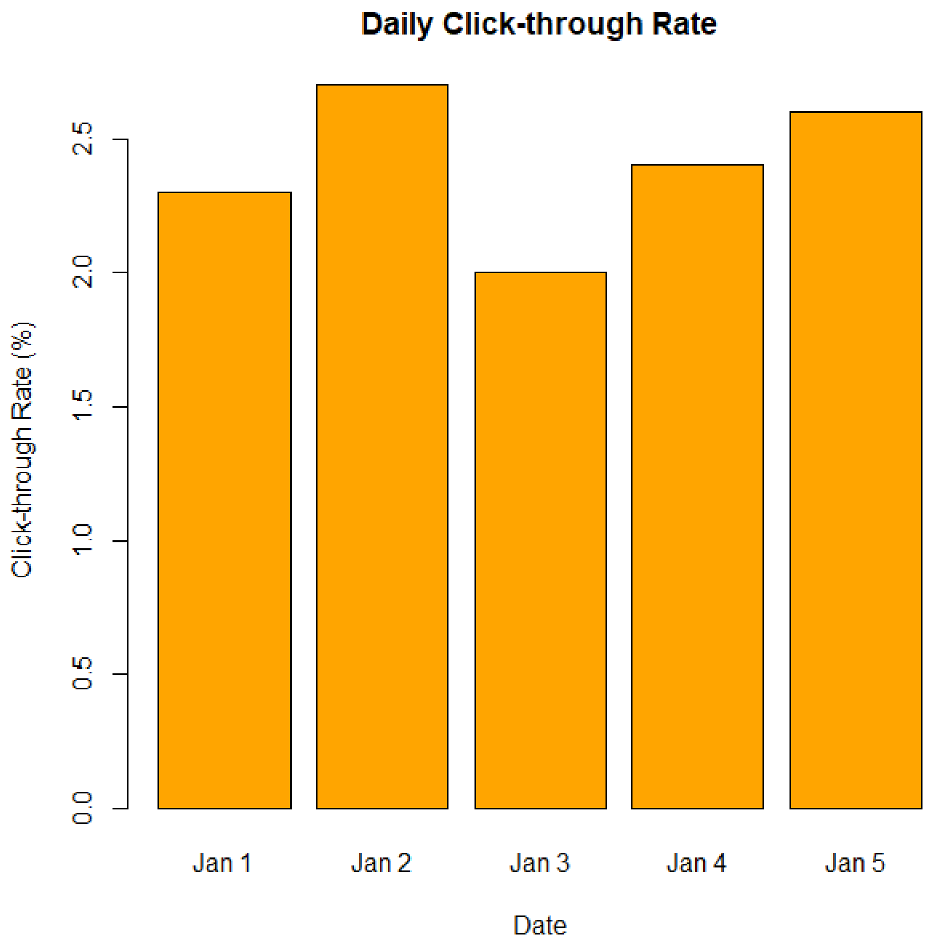
Code:

```
Date <- c("Jan 1","Jan 2","Jan 3","Jan 4","Jan 5")

CTR <- c(2.3,2.7,2.0,2.4,2.6)

barplot(
CTR,
names.arg=Date,
col="orange",
xlab="Date",
ylab="Click-through Rate (%)",
main="Daily Click-through Rate"
)
```

Output:



3. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

Code:

```
Days <- 1:5

Likes <- c(100,120,140,160,180)
Shares <- c(50,60,70,80,90)
Comments <- c(20,25,30,35,40)

interaction_data <- rbind(Likes,Shares,Comments)

barplot(
interaction_data,
col=c("skyblue","orange","lightgreen"),
main="User Interactions",
xlab="Days",
ylab="Count",
legend.text=c("Likes","Shares","Comments")
)
```

Output: